



Automotive CAN Bus Data Logger with Wi-Fi Upload

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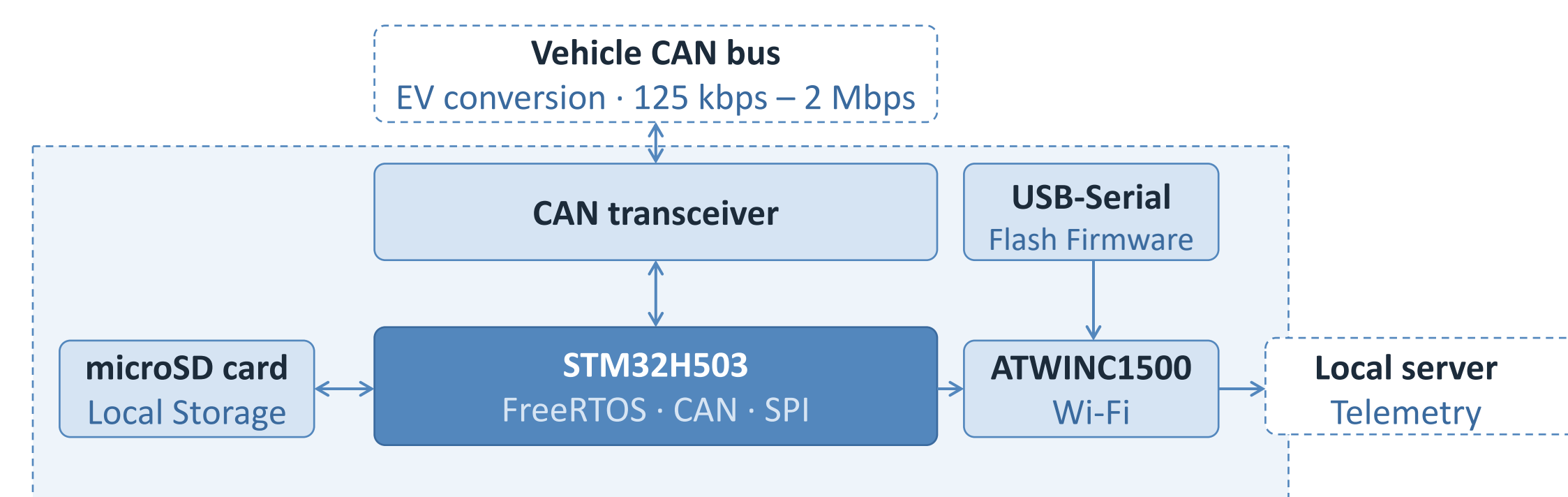


MAKE WAVES.

Abstract

A low-cost automotive CAN Bus data logger built for Peace Vans, a Seattle-based EV conversion company. The device captures CAN frames to a microSD card and automatically uploads data via Wi-Fi when in range of a configured network during charging. Built on the STM32H503 under FreeRTOS, a companion Python app decodes logs using DBC files, enabling remote fault diagnosis without traveling to the customer.

System Block Diagram

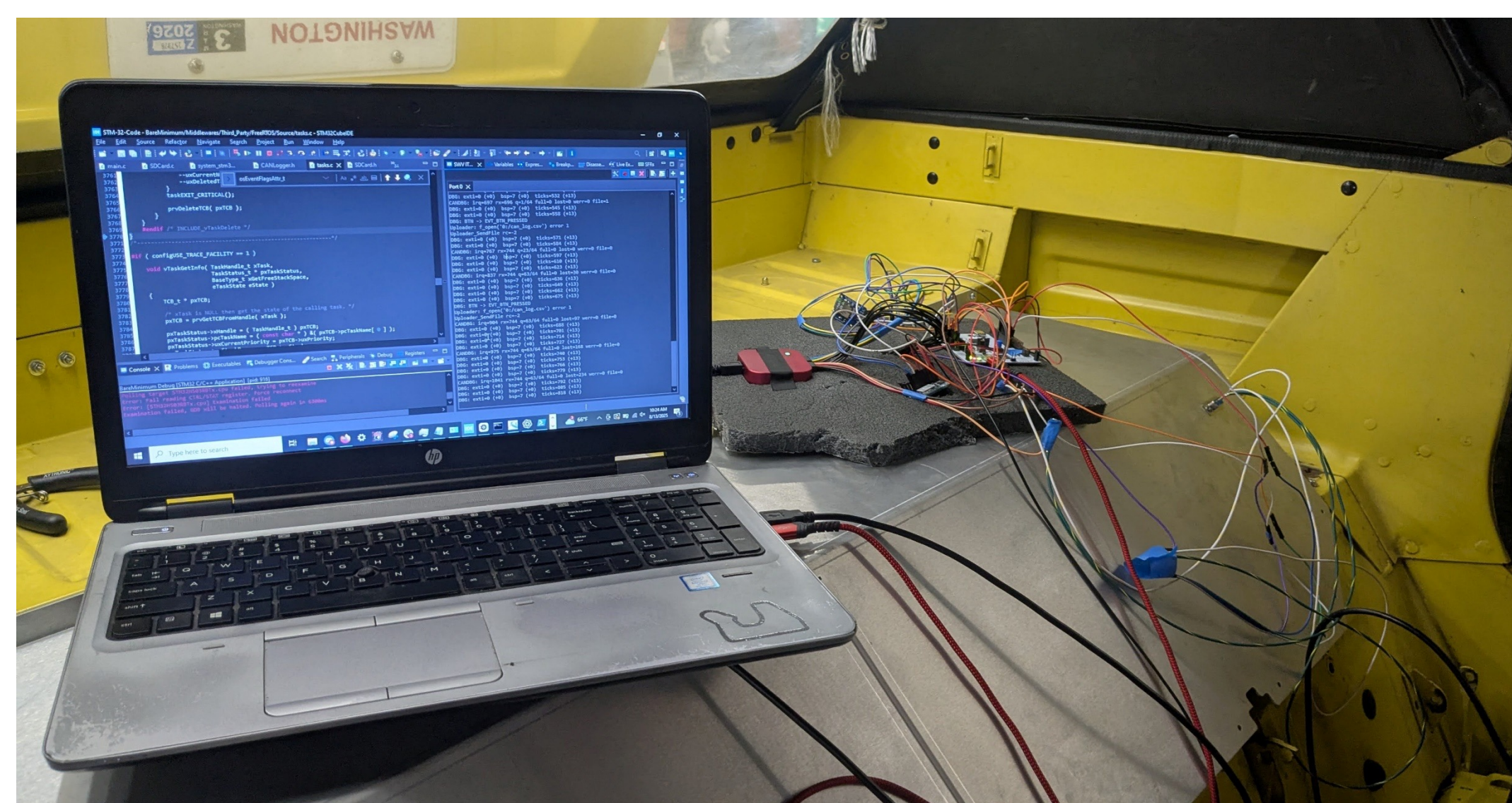


Background

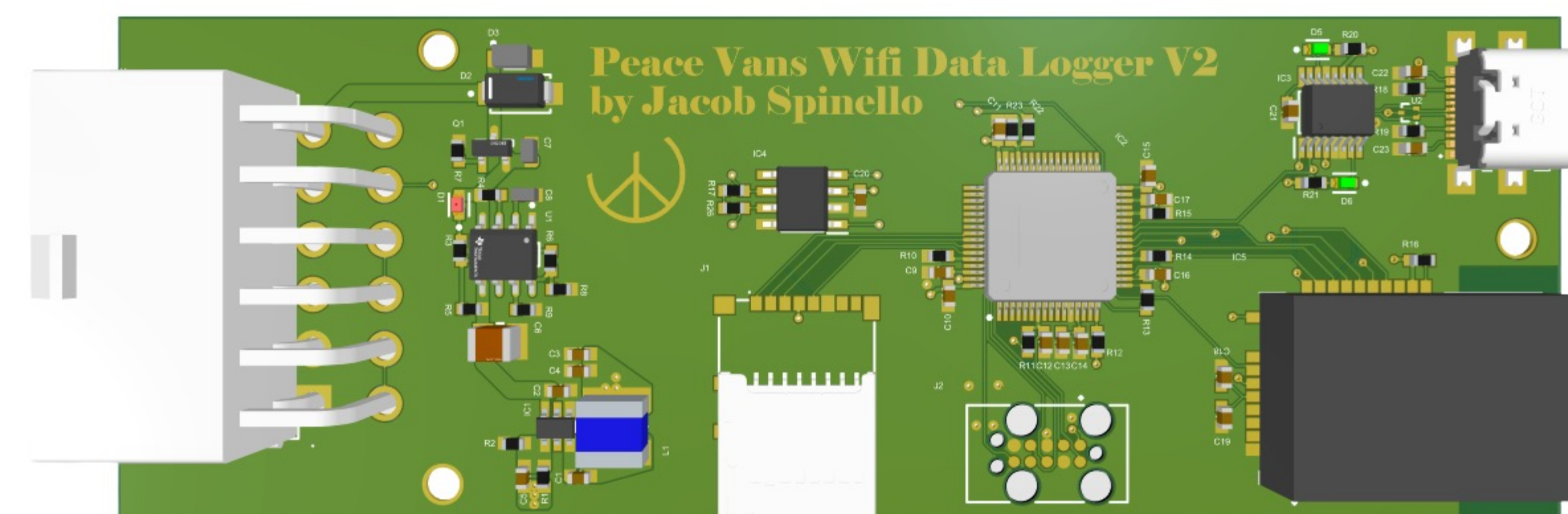
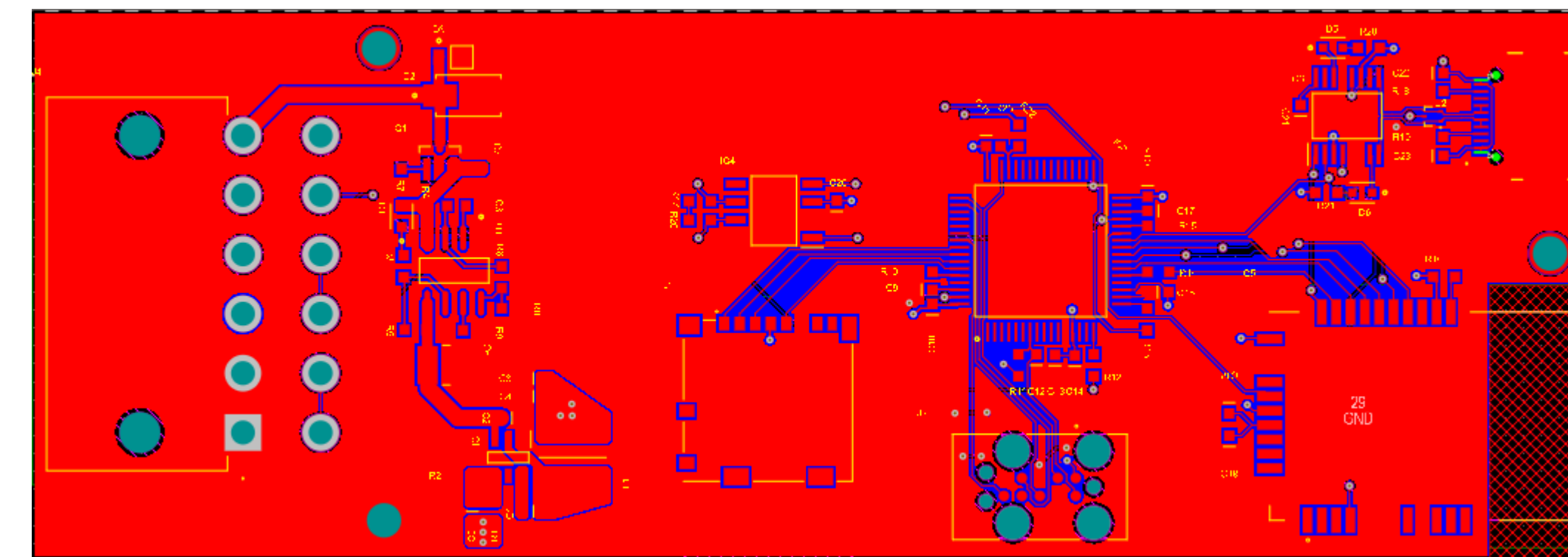
Peace Vans is a Seattle-based EV conversion shop whose vehicles are highly customized systems. When a customer vehicle faults remotely, there is no good way to diagnose it. The current market leader, the Orion BMS Connect, only provides high-level pack data and requires an active Wi-Fi connection, meaning nothing is logged while driving. For a shop needing full vehicle parameters captured in the field, it falls far short.

This project fills that gap with a logger that captures the full CAN bus during every drive and uploads via Wi-Fi, allowing Peace Vans to remotely diagnose a vehicle anywhere in the country without sending an engineer on site.

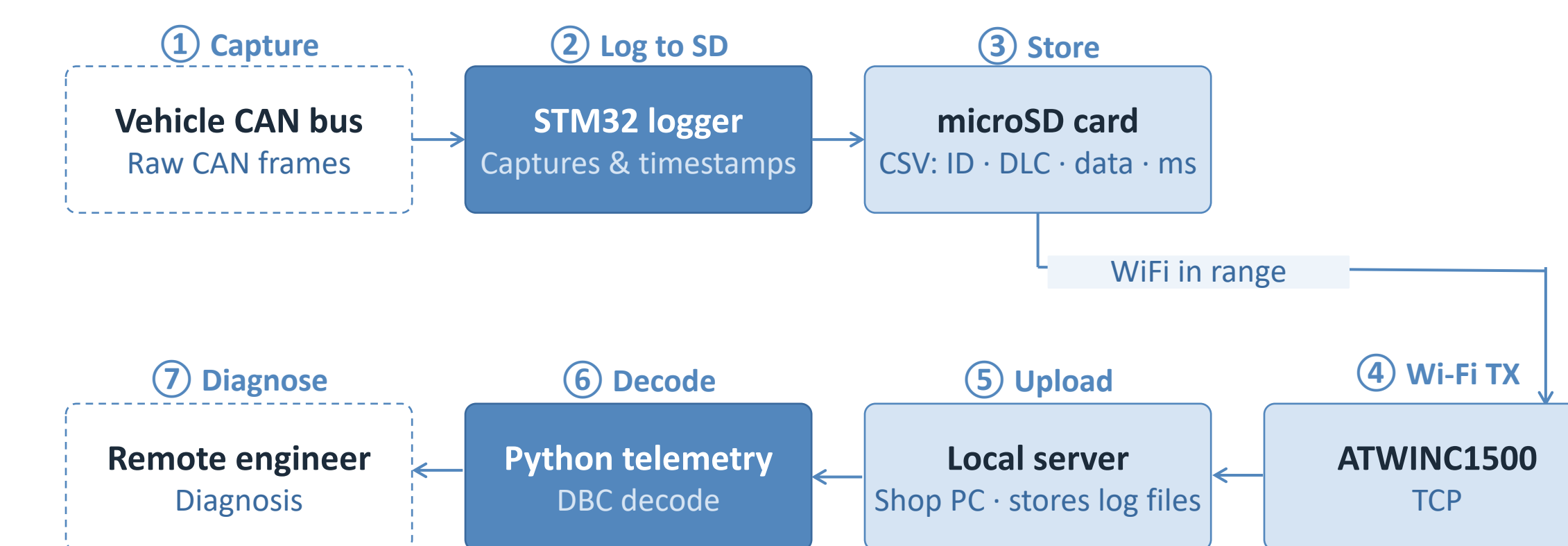
Initial Prototype



Altium PCB Design



Data Flow Diagram



Revision

The two primary drivers for revision was inefficient power regulation and handling firmware updates for the Wi-Fi chip.

- The original LDO dissipated enough heat during extended logging to trigger thermal shutdown, resolved by switching to a buck converter for more efficient and consistent power delivery. Ripple was not a concern given the fully digital design.
- A USB-C to serial IC was added to the board after realizing ATWINC1500 modules sourced from different distributors may ship with varying firmware versions. Rather than updating the codebase to match every revision, any module can be flashed to a known-good firmware version before installation without modifying the main codebase.

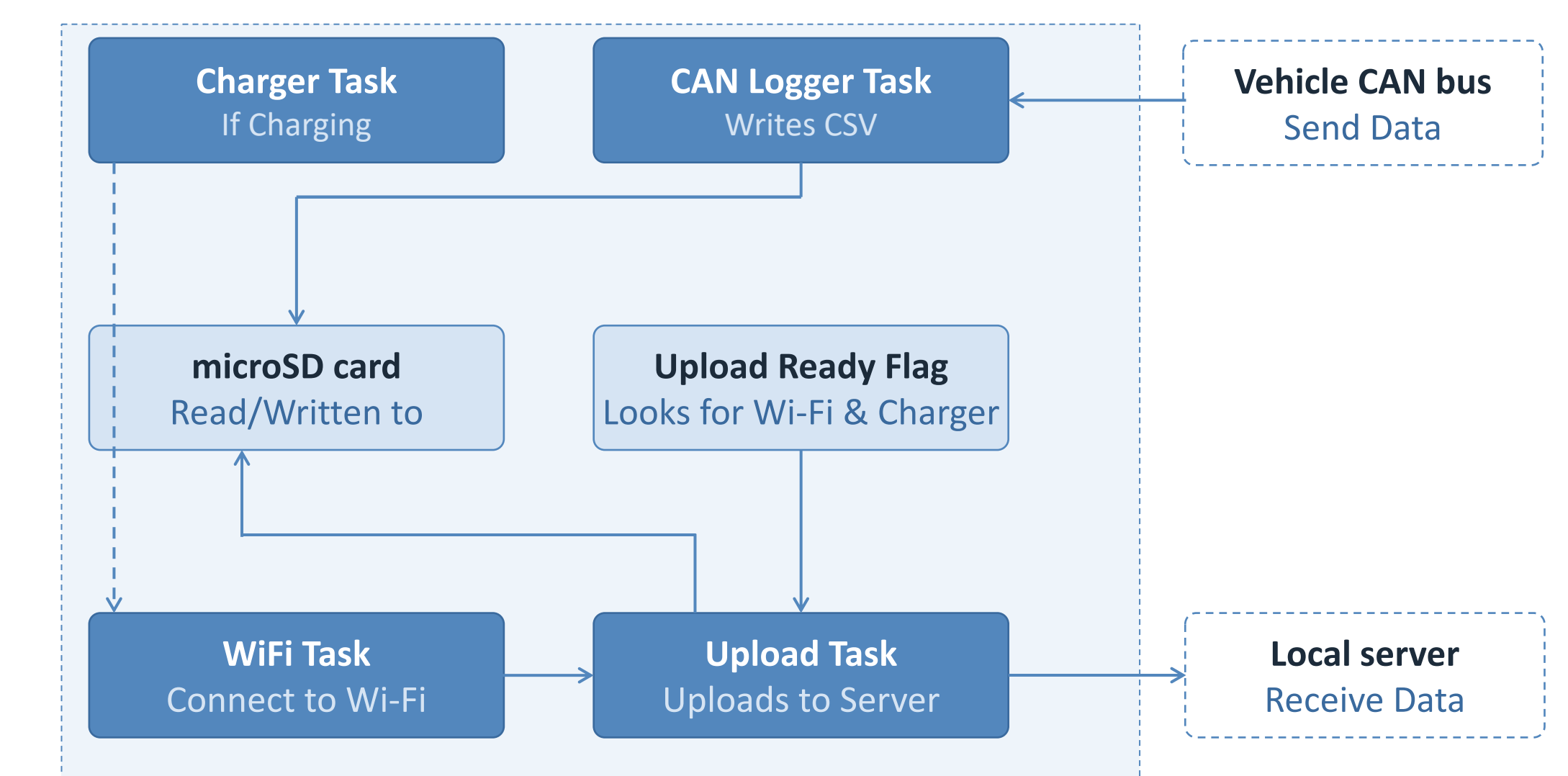
Python Telemetry App



The goal of this app is to make the data from the system easy to view and understand. Some features include:

- Accepts a DBC to decode raw CAN Values into a human readable format
- Telemetry style plotting
- Parse data to find minimum, maximum, or average values over time

RTOS Structure



Future Direction

Wi-Fi 6 support: Replace the ATWINC1500 with a module supporting 802.11ax to ensure compatibility with modern mobile hotspots and infrastructure
Analog Sensor Support: Enable user to configure extra sensors in the system that may not appear on the CAN Bus

Acknowledgements

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